

Symmetry breaking colimit of topological phases and equivariant coarse indices

joint work with Guo Chuan Thiang, [arxiv:2607.28181](https://arxiv.org/abs/2607.28181).



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Goal of my talk

Let G be a discrete countable group.

We shall provide a dichotomy of the $K(O)$ -theory of the reduced group C^* -algebra / equivariant Roe C^* -algebra of G into **strong/weak** parts:

$$0 \rightarrow wK_*(C^*(X)^G) \rightarrow K_*(C^*(X)^G) \rightarrow sK_*(C^*(X)^G) \rightarrow 0.$$


This is motivated by the notion of strong/weak phases of topological insulators and their macroscopic nature.

A geometric setup for higher/coarse index

Let:

- ▶ X be a Riemannian manifold, $\Gamma \curvearrowright X$ discrete, proper, cocompact.
- ▶ $\mathcal{D}$ be a Γ -invariant self-adjoint Dirac-type operator.

The Γ -equivariant (coarse/higher) index of $\mathcal{D}$ is defined as a certain K-theory class

$$\mathrm{ind}_\Gamma(\mathcal{D}) \in K_*(C^*(X)^\Gamma).$$

- ▶ $C^*(X)^\Gamma$: equivariant Roe algebra.
- ▶ Example: M a complete Riemannian spin manifold, $X := \tilde{M}$ the universal cover, $\Gamma := \pi_1(X)$, $\mathcal{D}$ the lifted Dirac operator on X .

Equivariant Roe algebra, 1

Let Γ be a discrete countable group, $\Gamma \curvearrowright X$ proper action.

A (twisted) X - Γ -module is a Hilbert space $\mathcal{H}$ equipped with:

- ▶ a non-degenerate $*$ -representation $\varrho: C_0(X) \rightarrow \mathbb{B}(\mathcal{H})$;
- ▶ a (σ -twisted) unitary representation $U: \Gamma \rightarrow \text{PU}(\mathcal{H})$;
- ▶ covariant: $U_g \varrho(f) U_g^* = \varrho(\alpha_g(f))$.

Definition

The **equivariant Roe algebra** $C^*(X, \mathcal{H})^{\Gamma, \sigma}$ is the C^* -subalgebra of $\mathbb{B}(\mathcal{H})$ generated by all Γ -invariant, finite propagation and locally compact operators.

Equivariant Roe algebra, 2

Basic Setup

- X proper metric space.
- Γ discrete countable group, acting properly on X .
- $\mathcal{H}$ σ -twisted X - Γ -module admitting a **bounded** fundamental domain.

Proposition Under the **basic setup**:

$$C^*(X, \mathcal{H})^{\Gamma, \sigma} \simeq C_\rho^*(\Gamma, \overline{\sigma}) \otimes \mathbb{K}(\mathcal{H}_D)$$

- ▶ $\mathcal{H}_D := \varrho(\chi_D)(\mathcal{H})$ for D a **fundamental domain** for $\Gamma \curvearrowright X$.
- ▶ $C_\rho^*(\Gamma, \overline{\sigma})$: twisted reduced group C*-algebra generated by right regular representation.

Consequence: $C^*(X, \mathcal{H})^{\Gamma, \sigma}$ is independent of X or $\mathcal{H}$ up to Morita equivalence.

An abelian/physical example

- ▶ Let $X = \mathbb{R}^d$, $\Gamma \simeq \mathbb{Z}^d$ a lattice of X .
- ▶ Consider the following Schrödinger operator on $L^2(\mathbb{R}^d)$:

$$H := -\Delta + V, \quad \text{with } V \in L^\infty(\mathbb{R}^d) \text{ being } \Gamma\text{-invariant.}$$

- ▶ Assume that the “Fermi energy” $E \notin \text{spec}(H)$. Then

$$P_{\leq E} := \chi_{(-\infty, E)}(H) \in C^*(\mathbb{R}^d, L^2\mathbb{R}^d)^\Gamma$$

and defines a class in $K_0(C^*(\mathbb{R}^d, L^2\mathbb{R}^d)^\Gamma) \simeq K_0(C(\hat{\Gamma})) \simeq \mathbb{Z}^{2^{d-1}}$.

- ▶ $d = 2$:

$$K_0(C(\mathbb{T}^2)) \simeq \underbrace{\mathbb{Z}}_{\text{rank of v.b.}} \oplus \underbrace{\mathbb{Z}}_{\text{Chern number}}.$$

- ▶ Physically: Chern number of $P_{\leq E}$ determines the **topological phase** of the material.

Strong/weak topological phases

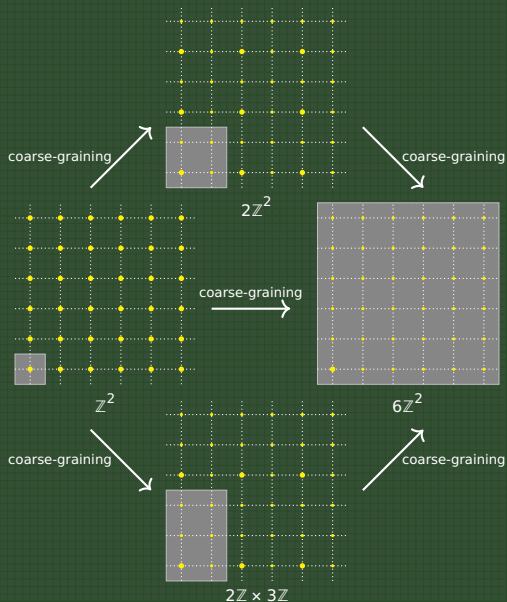
More interesting topological phases/invariants occur in **KO**-theory. They can be **torsion**, and can be distinguished by weak/strong.

- ▶ The **strong** one is well-established experimentally.
- ▶ The **weak** ones are generally considered to be non-robust.

$$\begin{aligned} \mathrm{KO}_4(C^*(\mathbb{R}^2)^{\mathbb{Z}^2}) &\simeq \underbrace{\mathbb{Z}}_{\text{rank}} \oplus \underbrace{\mathbb{Z}/2}_{\text{Kane-Mele}}, \\ \mathrm{KO}_4(C^*(\mathbb{R}^3)^{\mathbb{Z}^3}) &\simeq \underbrace{\mathbb{Z}}_{\text{rank}} \oplus \underbrace{(\mathbb{Z}/2)^3}_{\text{weak}} \oplus \underbrace{\mathbb{Z}/2}_{\text{strong}}. \end{aligned}$$

Goal Find an “intrinsic” way to distinguish weak/strong phases, i.e. K-theory of equivariant Roe algebras/group C*-algebras.

Consistent symmetry breaking, 1



Consistent symmetry breaking, 2

- ▶ If $G \subseteq \Gamma$ is a subgroup, then there is a natural injective *-homomorphism

$$l_{G \setminus \Gamma}: C^*(X, \mathcal{H})^{\Gamma, \sigma} \hookrightarrow C^*(X, \mathcal{H})^{G, \sigma}.$$

by viewing $\mathcal{H}$ as an **X - G -module**
and restricting the group 2-cocycle $\sigma: \Gamma \times \Gamma \rightarrow U(1)$ to G .

- ▶ Such injective maps are consistently related:

$$l_{K \setminus G} = l_{K \setminus H} \circ l_{H \setminus G}, \quad K \subseteq H \subseteq G.$$

- ▶ This gives a **directed system** of equivariant Roe algebras and their K-theory.

Symmetry breaking Roe algebra

Definition The symmetry breaking (directed) poset is

$$\mathfrak{S} = \{\text{finite-index subgroups of } \Gamma\}$$

and $G \prec H$ iff H is a subgroup of G .

The symmetry breaking Roe algebra

$$C^*(X, \mathcal{H})^{\mathfrak{S}} := \varinjlim_{G \in \mathfrak{S}} C^*(X, \mathcal{H})^G \simeq \overline{\bigcup_{G \in \mathfrak{S}} C^*(X, \mathcal{H})^G}.$$

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Consistency with symmetry breaking means that any G -dependent physical prediction must factor through this colimit:

A commutative diagram illustrating the consistency of physical predictions with symmetry breaking. The diagram consists of three nodes and three arrows:

- Top-left node: $K_*(C^*(X, \mathcal{H})^G)$
- Bottom-center node: $K_*(C^*(X, \mathcal{H})^{\mathfrak{S}})$
- Top-right node: experimental quantities

The arrows are:

- A solid horizontal arrow from $K_*(C^*(X, \mathcal{H})^G)$ to experimental quantities.
- A solid diagonal arrow from $K_*(C^*(X, \mathcal{H})^G)$ to $K_*(C^*(X, \mathcal{H})^{\mathfrak{S}})$, labeled $\iota_{\mathfrak{S} \setminus G}$.
- A dotted diagonal arrow from $K_*(C^*(X, \mathcal{H})^{\mathfrak{S}})$ to experimental quantities.

Symmetry breaking in K-theory, 1

- Let $H \subseteq G$. The symmetry breaking maps

$$l_{H \setminus G}: C^*(X, \mathcal{H})^{G, \sigma} \rightarrow C^*(X, \mathcal{H})^{H, \sigma}$$

induces a *-homomorphism $C_\rho^*(G, \bar{\sigma}) \rightarrow \mathbb{M}_{[G:H]} C_\rho^*(H, \bar{\sigma})$.

- Such induced *-homomorphism is **not** canonical: it depends on the split section $H \setminus G \rightarrow G$ and the fundamental domains for G and H .
- Nevertheless: the induced **K-theory** maps are well-defined and gives an isomorphic **directed system of abelian groups**:

$$\begin{array}{ccc} K_i(C^*(X, \mathcal{H})^G) & \xrightarrow{K_i(l_{H \setminus G})} & K_i(C^*(X, \mathcal{H})^H) \\ \alpha_G \downarrow \sim & & \sim \downarrow \alpha_H \\ K_i(C_\rho^*(G, \bar{\sigma})) & \xrightarrow{\phi_{H \setminus G}^{\bar{\sigma}}} & K_i(C_\rho^*(H, \bar{\sigma})). \end{array}$$

Symmetry breaking in K-theory, 2

Example Let $X = \mathbb{R}$, $\Gamma \simeq \mathbb{Z}$.

- ▶ Symmetry breaking poset $\mathfrak{S} = \{m\mathbb{Z} \mid m \in \mathbb{N}\}$.
- ▶ The “symmetry breaking” map

$$K_i(\iota_{n \setminus m}): K_i(C^*(\mathbb{R})^{m\mathbb{Z}}) \rightarrow K_i(C^*(\mathbb{R})^{n\mathbb{Z}}), \quad m \mid n.$$

can be identified with the induced map of the *-homomorphism

$$C_\rho^*(m\mathbb{Z}) \rightarrow C_\rho^*(n\mathbb{Z}) \otimes \mathbb{K}(\ell^2(n\mathbb{Z} \setminus m\mathbb{Z})) \simeq \mathbb{M}_N(C^*(n\mathbb{Z})), \quad N := \frac{n}{m}.$$

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- $i = 0: [1] \mapsto \frac{n}{m}[1]$.
- $i = 1: [w_m] \mapsto \left[\begin{pmatrix} 0 & \text{id}_N \\ w_n & 0 \end{pmatrix} \right] = [w_n].$

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- $i = 1: [w_m] \mapsto \left[\begin{pmatrix} 0 & \text{id}_N \\ w_n & 0 \end{pmatrix} \right] = [w_n]$.
- ▶ Passing to the (directed) colimit:

$$K_i(C^*(\mathbb{R})^{\mathfrak{S}}) \simeq \begin{cases} \mathbb{Q} & i = 0; \\ \mathbb{Z} & i = 1. \end{cases}$$

Symmetry breaking in K-theory, 3

Strategies in higher-dimensional / real / twisted cases

- ▶ In higher dimensions $X = \mathbb{R}^d$ and $\Gamma \simeq \mathbb{Z}^d$: passing to a **cofinal** subset and apply Schochet's Künneth Theorem.
- ▶ The **real** case: similar with Boersema's Künneth Theorem.
- ▶ The **twisted** case: no Künneth formula. Using iterated **Pimsner-Voiculescu sequence**, together with the **strong-to-coarse comparison maps**.

Symmetry breaking in K-theory, 3

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Results of physical relevance

$$K_0(C^*(\mathbb{R}^2)^\mathbb{G}) \simeq \mathbb{Q} \oplus \mathbb{Z} \quad (\text{Chern number})$$

$$KO_4(C^*(\mathbb{R}^2)^\mathbb{G}) \simeq \mathbb{Q} \oplus \mathbb{Z}/2 \quad (\text{Kane-Mele invariant})$$

$$KO_4(C^*(\mathbb{R}^3)^\mathbb{G}) \simeq \mathbb{Q} \oplus \mathbb{Z}/2 \quad (\text{strong } \mathbb{Z}/2\text{-invariant}).$$

Symmetry breaking in K-theory, 4

Theorem $K(O)$ -theory of $C^*(\mathbb{R}^d)^\mathbb{G}$ for a twisted/real $\mathbb{R}^d$ - $\mathbb{Z}^d$ -module:

$d \backslash i$	0	1	2	3	4	5	6	7	8
K_0	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{Q} \oplus \mathbb{Z}$	$\mathbb{Q}^4$	$\mathbb{Q}^7 \oplus \mathbb{Z}$	$\mathbb{Q}^{16}$	$\mathbb{Q}^{31} \oplus \mathbb{Z}$	$\mathbb{Q}^{64}$	$\mathbb{Q}^{127} \oplus \mathbb{Z}$
K_1	0	$\mathbb{Z}$	$\mathbb{Q}^2$	$\mathbb{Q}^3 \oplus \mathbb{Z}$	$\mathbb{Q}^8$	$\mathbb{Q}^{15} \oplus \mathbb{Z}$	$\mathbb{Q}^{32}$	$\mathbb{Q}^{63} \oplus \mathbb{Z}$	$\mathbb{Q}^{128}$

$d \backslash i$	0	1	2	3	4	5	6	7	8
KO_0	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{Q}$	$\mathbb{Q}$	$\mathbb{Q} \oplus \mathbb{Z}$	$\mathbb{Q}^6$	$\mathbb{Q}^{16} \oplus \mathbb{Z}/2$	$\mathbb{Q}^{36} \oplus \mathbb{Z}/2$	$\mathbb{Q}^{71} \oplus \mathbb{Z}$
KO_1	$\mathbb{Z}/2$	$\mathbb{Z}$	$\mathbb{Q}^2$	$\mathbb{Q}^3$	$\mathbb{Q}^4$	$\mathbb{Q}^5 \oplus \mathbb{Z}$	$\mathbb{Q}^{12}$	$\mathbb{Q}^{28} \oplus \mathbb{Z}/2$	$\mathbb{Q}^{64} \oplus \mathbb{Z}/2$
KO_2	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}$	$\mathbb{Q}^3$	$\mathbb{Q}^6$	$\mathbb{Q}^{10}$	$\mathbb{Q}^{15} \oplus \mathbb{Z}$	$\mathbb{Q}^{28}$	$\mathbb{Q}^{56} \oplus \mathbb{Z}/2$
KO_3	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}$	$\mathbb{Q}^4$	$\mathbb{Q}^{10}$	$\mathbb{Q}^{20}$	$\mathbb{Q}^{35} \oplus \mathbb{Z}$	$\mathbb{Q}^{64}$
KO_4	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{Q} \oplus \mathbb{Z}/2$	$\mathbb{Q} \oplus \mathbb{Z}/2$	$\mathbb{Q} \oplus \mathbb{Z}$	$\mathbb{Q}^6$	$\mathbb{Q}^{16}$	$\mathbb{Q}^{36}$	$\mathbb{Q}^{70} \oplus \mathbb{Z}$
KO_5	0	$\mathbb{Z}$	$\mathbb{Q}^2$	$\mathbb{Q}^3 \oplus \mathbb{Z}/2$	$\mathbb{Q}^4 \oplus \mathbb{Z}/2$	$\mathbb{Q}^5 \oplus \mathbb{Z}$	$\mathbb{Q}^{12}$	$\mathbb{Q}^{28}$	$\mathbb{Q}^{64}$
KO_6	0	0	$\mathbb{Z}$	$\mathbb{Q}^3$	$\mathbb{Q}^6 \oplus \mathbb{Z}/2$	$\mathbb{Q}^{10} \oplus \mathbb{Z}/2$	$\mathbb{Q}^{15} \oplus \mathbb{Z}$	$\mathbb{Q}^{28}$	$\mathbb{Q}^{56}$
KO_7	0	0	0	$\mathbb{Z}$	$\mathbb{Q}^4$	$\mathbb{Q}^{10} \oplus \mathbb{Z}/2$	$\mathbb{Q}^{20} \oplus \mathbb{Z}/2$	$\mathbb{Q}^{35} \oplus \mathbb{Z}$	$\mathbb{Q}^{64}$

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KO_1	$\mathbb{Z}/2$	$\mathbb{Z}$	$\mathbb{Q}^2$	$\mathbb{Q}^3$	$\mathbb{Q}^4$	$\mathbb{Q}^5 \oplus \mathbb{Z}$	$\mathbb{Q}^{12}$	$\mathbb{Q}^{28} \oplus \mathbb{Z}/2$	$\mathbb{Q}^{64} \oplus \mathbb{Z}/2$
KO_2	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}$	$\mathbb{Q}^3$	$\mathbb{Q}^6$	$\mathbb{Q}^{10}$	$\mathbb{Q}^{15} \oplus \mathbb{Z}$	$\mathbb{Q}^{28}$	$\mathbb{Q}^{56} \oplus \mathbb{Z}/2$
KO_3	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}$	$\mathbb{Q}^4$	$\mathbb{Q}^{10}$	$\mathbb{Q}^{20}$	$\mathbb{Q}^{35} \oplus \mathbb{Z}$	$\mathbb{Q}^{64}$
KO_4	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{Q} \oplus \mathbb{Z}/2$	$\mathbb{Q} \oplus \mathbb{Z}/2$	$\mathbb{Q} \oplus \mathbb{Z}$	$\mathbb{Q}^6$	$\mathbb{Q}^{16}$	$\mathbb{Q}^{36}$	$\mathbb{Q}^{70} \oplus \mathbb{Z}$
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A dichotomy for weak/strong K-theory, 1

The **reduced** part of $K(O)$ -theory of $C^*(\mathbb{R}^d)^G$:

$i \backslash d$	0	1	2	3	4	5	6	7	8
sK_0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$
sK_1	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0

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sKO_0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}$
sKO_1	$\mathbb{Z}/2$	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$
sKO_2	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$	0	$\mathbb{Z}/2$
sKO_3	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$	0
sKO_4	$\mathbb{Z}$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$
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sKO_6	0	0	$\mathbb{Z}$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}$	0	0
sKO_7	0	0	0	$\mathbb{Z}$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}$	0

This is the same as Kitaev's periodic table / homotopy groups of $U(\infty)$ or $O(\infty)$ / $K(O)$ -theory of Roe algebra.

A dichotomy for weak/strong K-theory, 2

Definition

- The **weak (symmetry breaking) colimit**

$$\mathrm{w}K_i(C^*(X)^{\mathfrak{S}}) := \text{maximal divisible subgroup of } K_i(C^*(X)^{\mathfrak{S}})$$

- The **strong (symmetry breaking) colimit**

$$\mathrm{s}K_i(C^*(X)^{\mathfrak{S}}) := K_i(C^*(X)^{\mathfrak{S}}) / \mathrm{w}K_i(C^*(X)^{\mathfrak{S}})$$

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For each $G \in \mathfrak{S}$: We have universal map $\iota_{\mathfrak{S} \setminus G} : C^*(X)^G \hookrightarrow C^*(X)^{\mathfrak{S}}$.

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For each $G \in \mathfrak{S}$: We have universal map $\iota_{\mathfrak{S} \setminus G} : C^*(X)^G \hookrightarrow C^*(X)^{\mathfrak{S}}$.

- The **weak subgroup**

$$\mathrm{w}K_i(C^*(X)^G) := K_i(\iota_{\mathfrak{S} \setminus G})^{-1}(\mathrm{w}K_i(C^*(X)^{\mathfrak{S}}))$$

- The **strong quotient**

$$\mathrm{s}K_i(C^*(X)^G) := K_i(C^*(X)^G) / \mathrm{w}K_i(C^*(X)^G)$$

Universal property of strong K-theory, 1

- For every abelian group A , let W be its maximal divisible subgroup. Then there is a short exact sequence which splits non-canonically:

$$0 \rightarrow W \xrightarrow{i} A \xrightarrow[\text{f...}]{q} A/W \rightarrow 0.$$

Universal property of strong K-theory, 1

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$$0 \rightarrow W \xrightarrow{i} A \xrightarrow[q, \dots]{q} A/W \rightarrow 0.$$

- Let R be a reduced (i.e. possesses a trivial maximal divisible subgroup) abelian group. Then any group homomorphism $A \rightarrow R$ factors:

$$\begin{array}{ccc} A & \xrightarrow{\quad} & R \\ & \searrow q & \nearrow \text{dotted} \\ & A/W & \end{array}$$

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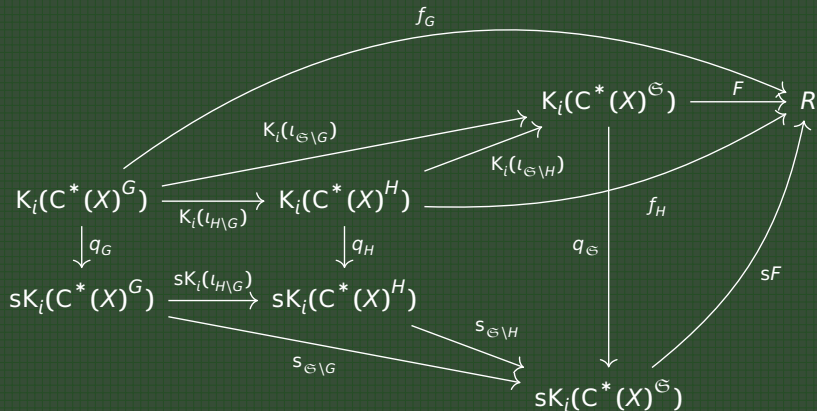
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- Assume **consistency with symmetry breaking**:
If R is a receptacle of **quantised** invariants, then $K_i(C^*(X)^G) \rightarrow R$ must factor through $sK_i(C^*(X)^G) \rightarrow R$.

Universal property of strong K-theory, 2



Strong-to-coarse comparison maps, 1

- A useful receptacle reduced abelian group R :
K(O)-theory of **non-equivariant Roe algebras**.

$$K_*(C^*(\mathbb{R}^d)) \simeq K_{*-d}(\mathbb{C}), \quad KO_*(C^*(\mathbb{R}^d)) \simeq KO_{*-d}(\mathbb{R}).$$

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- Forgetting the equivariance altogether gives the “strong-to-coarse comparison map” for each $G \in \mathfrak{S}$:

$$sK_i(l_{1 \setminus G}): sK_i(C^*(X, \mathcal{H})^G) \rightarrow K_i(C^*(X, \mathcal{H}))$$

Strong-to-coarse comparison maps, 1

- ▶ A useful receptacle reduced abelian group R : $K(O)$ -theory of **non-equivariant Roe algebras**.

$$K_*(C^*(\mathbb{R}^d)) \simeq K_{*-d}(\mathbb{C}), \quad KO_*(C^*(\mathbb{R}^d)) \simeq KO_{*-d}(\mathbb{R}).$$

- ▶ Forgetting the equivariance altogether gives the “strong-to-coarse comparison map” for each $G \in \mathfrak{S}$:

$$sK_i(\iota_{1 \setminus G}): sK_i(C^*(X, \mathcal{H})^G) \rightarrow K_i(C^*(X, \mathcal{H}))$$

- ▶ The map $sK_i(\iota_{1 \setminus G})$ can be used to detect non-trivial elements in the strong K -theory of $C^*(X, \mathcal{H})^G$ (equivalently, $C_\rho^*(G)$).

Strong-to-coarse comparison maps, 2

Theorem Let

Γ finitely generated, torsion free, nilpotent discrete group.

$\mathcal{H}$ ample, σ -twisted complex X - Γ -module
satisfying the basic setup.

σ magnetic 2-cocycle for $(\Gamma_{\mathbb{R}}, \Gamma)$
($\Gamma_{\mathbb{R}}$: Mal'cev completion of Γ).

Then for every $G \in \mathfrak{S}$, the strong-to-coarse comparison maps

$$sK_i(C^*(X, \mathcal{H})^G) \rightarrow K_i(C^*(X, \mathcal{H}))$$

are surjective.

Strong-to-coarse comparison maps, 2

Theorem Let

Γ finitely generated, torsion free, nilpotent discrete group.

$\mathcal{H}$ ample, σ -twisted complex X - Γ -module
satisfying the basic setup.

σ magnetic 2-cocycle for $(\Gamma_{\mathbb{R}}, \Gamma)$
($\Gamma_{\mathbb{R}}$: Mal'cev completion of Γ).

Then for every $G \in \mathfrak{S}$, the strong-to-coarse comparison maps

$$sK_i(C^*(X, \mathcal{H})^G) \rightarrow K_i(C^*(X, \mathcal{H}))$$

are surjective.

Proof. The generators of $K_i(C^*(\Gamma_{\mathbb{R}}, \mathcal{H}))$ is the (coarse) index class of the spin Dirac operator on $\Gamma_{\mathbb{R}}$. It is G -invariant for every G (using contractibility of $\Gamma_{\mathbb{R}}$). So it lifts to every $sK_i(C^*(\Gamma_{\mathbb{R}}, \mathcal{H})^G)$. □

Strong-to-coarse comparison maps, 3

Remark

1. “Magnetic twists” are allowed, which gives rise to “magnetic translations” commuting with the twisted Dirac operator.
2. Same holds if $\Gamma_{\mathbb{R}}$ is replaced by any Lie group Y containing Γ as a lattice, and such that the **coarse Baum–Connes assembly map** is an isomorphism, e.g. surface groups.
3. Same holds in KO-theory for untwisted free abelian Γ .
The generator of both “strong”/“coarse” K-theory is the Hodge–de Rham operator on the “real” tori $\hat{\Gamma}$.

Application: TKNN–Chern number of Landau levels

- Consider the **twisted** Dirac operator on $L^2(\mathbb{R}^2, \mathbb{C}^2)$:

$$\mathcal{D}_b = \begin{pmatrix} 0 & \mathcal{D}_b^- \\ \mathcal{D}_b^+ & 0 \end{pmatrix} = -2i \begin{pmatrix} 0 & \partial - \frac{b}{4}\bar{z} \\ \bar{\partial} + \frac{b}{4}z & 0 \end{pmatrix}.$$

- Lowest Landau level** $:= \ker \mathcal{D}_b^+ = \overline{\text{span}} \left\{ z \mapsto z^m e^{-b|z|^2/4} \right\}$.
- The eigenprojection onto LLL

$$[P_{\text{LLL},b}] \in K_0(C^*(\mathbb{R}^2))$$

is a generator.

- This forces the (TKNN–)Chern number of $P_{\text{LLL},b}$ to be ± 1 :

$$\underbrace{K_0(C^*(X)^G)}_{\mathbb{Z} \oplus \mathbb{Z}} \xrightarrow{q_G} \underbrace{SK_0(C^*(X)^G)}_{\mathbb{Z}} \xrightarrow[\sim]{SK_0(\iota_{1 \setminus G})} \underbrace{K_0(C^*(X))}_{\mathbb{Z}}$$

$$[P_{\text{LLL},b}]_G = (r, c) \longmapsto c \longmapsto [P_{\text{LLL},b}] = \pm 1.$$

Strong equals coarse?

Question Is every strong-to-coarse comparison map an isomorphism?

More-or-less: let $|\Gamma| := (\Gamma, \text{proper left-invariant metric})$.

Are the induced K-theory maps

$$K_i(C_\rho^*(\Gamma)) \rightarrow K_i(C^*(|\Gamma|))$$

always surjective?

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Idea Counterexamples may arise because:

- ▶ $K_i(C^*(|\Gamma|))$ is often singly generated by the “coarse-geometric fundamental class”.
- ▶ It lifts to the “(coarse) K-homological fundamental class” under the coarse Baum–Connes assembly map.
- ▶ It suffices to kill this class in $K_i(C_\rho^*(\Gamma))$, equivalently, topological K-theory of the group Γ .

A non-surjective example, 1

Example Let $\Gamma = \mathbb{Z} \rtimes \mathbb{Z}$ be the wallpaper group, i.e. the crystallographic group given by

$$\begin{array}{ccccccc} 1 & \longrightarrow & \mathbb{Z}^2 & \longrightarrow & \mathbb{Z} \rtimes \mathbb{Z} & \longrightarrow & \mathbb{Z}/2 \longrightarrow 1 \\ & & (a, b) & \longmapsto & (a, 2b) & & \\ & & & & (a, b) & \longmapsto & b \bmod 2\mathbb{Z}. \end{array}$$

Then:

- ▶ $B\Gamma = \text{Klein bottle } K$.
- ▶ The strong-to-coarse comparison maps

$$sK_i(C^*(K)^\Gamma) \rightarrow K_i(C^*(K))$$

vanishes in all degrees, whereas $K_i(C^*(K)) \simeq K_i(\mathbb{C})$.

- ▶ Proof by PV sequence.

A non-surjective example, 2

Notice that $\mathbb{Z} \rtimes \mathbb{Z}$ contains an abelian subgroup isomorphic to $\mathbb{Z}^2$. The strong-to-coarse comparison maps become isomorphisms there.

Theorem Let

Γ finitely generated, torsion free, nilpotent discrete group.

$\tilde{\Gamma}$ extension of a finite group F by Γ .

$\tilde{\mathfrak{S}}$ poset of finite-index subgroups of $\tilde{\Gamma}$.

$\mathcal{H}$ ample, σ -twisted complex X - $\tilde{\Gamma}$ -module.

σ 2-cocycle for $\tilde{\Gamma}$,

which restricts to a magnetic 2-cocycle for $(\Gamma_{\mathbb{R}}, \Gamma)$.

Then the strong-colimit-to-coarse comparison maps

$$sK_i(C^*(X, \mathcal{H})^{\tilde{\mathfrak{S}}}) \rightarrow K_i(C^*(X, \mathcal{H}))$$

are surjective.

Summary and outlook

- Instead of looking at $K_*(C^*(X)^\Gamma)$, it may be useful to look at the whole diagram

$$\{K_*(C^*(X)^G) \mid [\Gamma : G] < +\infty\}.$$

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“strong” equals “coarse”,
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- Instead of looking at $K_*(C^*(X)^\Gamma)$, it may be useful to look at the whole diagram

$$\{K_*(C^*(X)^G) \mid [\Gamma : G] < +\infty\}.$$

- This allows for a dichotomy of “strong/weak” K-theory elements by looking at the **divisibility** of their images in the colimit.
- In many cases (appearing naturally in physics!):
“strong” equals “coarse”,
but may go wrong already for crystallographic groups.
- Instead of looking at all finite-index subgroups, it is sometimes useful to look at smaller posets.
This allows for maintaining the presence of certain point groups.